

CSCE 649 PHYSICALLY-BASED MODELING · MODULE 07 · LECTURE NOTES

Rigid Body Dynamics

Rotation, which is where the interesting mistakes are.

MODULE OUTCOMES

Write the rigid body state and its derivative.

Compute and transform an inertia tensor.

Explain why angular momentum, not angular velocity, is the state variable.

Integrate orientation with quaternions and renormalise correctly.

Explain and reproduce the intermediate axis theorem.

1 State and equations of motion

Quantity	Symbol	Component s	Note
Position of the centre of mass	$\mathbf{x}$	3	—
Orientation	$\mathbf{q}$	4	A unit quaternion. §3.
Linear momentum	$\mathbf{P}$	3	$\mathbf{P} = m\mathbf{v}$.
Angular momentum	$\mathbf{L}$	3	Not angular velocity. This choice is the main content of the module.

$$\dot{\mathbf{x}} = \mathbf{P}/m, \quad \dot{\mathbf{P}} = \mathbf{F} \quad \dot{\mathbf{q}} = \frac{1}{2} \boldsymbol{\omega} \mathbf{q}, \quad \dot{\mathbf{L}} = \boldsymbol{\tau}$$

$$\boldsymbol{\omega} = \mathbf{I}^{-1}(\mathbf{t}) \mathbf{L}$$

The linear equations are exactly a particle's. The rotational ones mirror them: **torque changes angular momentum** just as force changes linear momentum. **Angular velocity is derived** from angular momentum through the current inertia tensor, and is not part of the state.

This is the line that matters. With no torque, $\mathbf{L}$ is constant — but $\boldsymbol{\omega}$ is not, because $\mathbf{I}$ depends on the body's current orientation and therefore changes as it rotates. A body under no torque at all can have a continuously changing angular velocity, which is the source of §4's behaviour.

2 The inertia tensor

DIRECTION-DEPENDENT MASS

Linear mass is a scalar: a body resists acceleration equally in every direction, so $F = ma$ with m a single number.

Rotational inertia is not a scalar. A long rod is easy to spin about its own axis and hard to spin end over end. The resistance depends on the axis.

So inertia is a 3×3 symmetric matrix, and $L = I\omega$ is a matrix equation. **The consequence is that angular momentum and angular velocity are generally not parallel** — a body can be spinning about one axis while its angular momentum points along another.

That non-parallelism is the origin of every interesting rotational phenomenon in rigid body dynamics: precession, nutation, gyroscopic resistance, and the intermediate axis instability of §4.

$$I_{\text{world}}(t) = R(t) I_{\text{body}} R^T(t) \quad I_{\text{world}}^{-1} = R I_{\text{body}}^{-1} R^T$$

The body-space tensor is a constant property of the shape and mass distribution. The world-space tensor changes continuously as the body rotates, and must be recomputed each step. **Store and transform the inverse**: every step needs I^{-1} to obtain ω from L , and nothing ever needs I itself.

- **Compute it from the volume, not the surface.** Integrating over the triangles of a mesh gives the inertia of a hollow shell, not a solid body. Mirtich's algorithm computes the volume integral exactly from the boundary representation and is the standard method.
- **Always about the centre of mass.** Compute the centre of mass first and translate the body so it lies at the origin, otherwise every equation in §1 is wrong.
- **Diagonalise at load time.** The eigenvectors of I are the **principal axes** and the eigenvalues the **principal moments**. Rotating the body into this frame makes I diagonal, which simplifies everything and is how you set up §4's demonstration.
- **Parallel axis theorem** for compound bodies: the inertia about any point equals the inertia about the centre of mass plus $m d^2$ terms. This is how you assemble a body from parts.
- **Verify against known shapes.** A solid sphere is $(2/5)mr^2$ about any axis; a solid box is $(m/12)(h^2 + d^2)$ about each principal axis. **An incorrect inertia tensor produces motion that looks subtly wrong with no identifiable error**, so check it directly rather than by eye.

3 Orientation

Representation	Number s	Problems
Euler angles	3	Gimbal lock — a degree of freedom is lost at certain orientations. The result depends on the order of the three rotations, and interpolation between two sets of angles does not follow a sensible path.
Rotation matrix	9	Numerical integration drifts away from orthonormality, so the body gradually shears and scales. Re-orthonormalising is possible (Gram–Schmidt) and awkward, and six redundant numbers is a lot of drift to control.
Quaternion	4	Must be renormalised every step. That is the entire disadvantage. One redundant number against three degrees of freedom, no singularities, and clean interpolation.

```

void RigidBody::step(double dt) {
    P += force * dt;
    x += (P / mass) * dt;

    L += torque * dt;                                // torque -> MOMENTUM
    mat3 R      = q.to_matrix();
    mat3 Iinv_w = R * Iinv_body * R.transpose();      // recompute from R
    vec3 omega  = Iinv_w * L;                          // DERIVE omega

    quat omega_q(0, omega.x, omega.y, omega.z);
    q = q + (0.5 * dt) * (omega_q * q);
    q.normalize();                                    // required
}

```

Three details to get right: torque integrates into $\mathbf{L}$ and not into ω ; the inverse inertia tensor is recomputed from the current rotation every step; and the quaternion is renormalised every step. **Omitting the normalisation is the classic bug** — the quaternion's magnitude drifts from 1, which the to-matrix conversion interprets as a uniform scale, and the body slowly grows or shrinks before degenerating entirely.

4 The intermediate axis theorem

THE DZHANIBEKOV EFFECT

A rigid body with three *distinct* principal moments of inertia — a T-handle, a tennis racket, a phone — spins stably about the axis of largest moment and about the axis of smallest moment.

Rotation about the intermediate axis is unstable. A body set spinning about it will flip end over end, periodically, indefinitely, **with no torque acting and nothing driving it**. The cosmonaut Vladimir Dzhanibekov famously filmed this with a wingnut in orbit.

Angular momentum is exactly conserved throughout the flip. Nothing is being added or removed. The motion is a consequence of the constraint that both $\mathbf{L}$ and the rotational kinetic energy are conserved, which confines ω to the intersection of two ellipsoids — and near the intermediate axis that intersection is a path that travels right round the body.

If your simulator reproduces this spontaneously, your rotational dynamics is correct. It is a far better test than watching things fall over, because nothing about the implementation hints at it — the behaviour has to emerge.

STORING ANGULAR VELOCITY IS WHAT BREAKS IT

With no torque, $\mathbf{L}$ is constant. ω is *not*, because $\omega = \mathbf{I}^{-1}(t)\mathbf{L}$ and $\mathbf{I}$ changes as the body rotates.

Store $\mathbf{L}$, recompute $\mathbf{I}^{-1}$ each step, and ω comes out correctly for free. The flip emerges from the equations without anything having been added.

Store ω and the equation of motion acquires an extra **gyroscopic term**, $\omega \times (\mathbf{I}\omega)$, which must be added explicitly. It is easy to omit, numerically awkward — it can destabilise an explicit integrator — and is frequently dropped from implementations for exactly that reason.

So storing momentum rather than velocity is not a stylistic preference. It is the difference between the correct behaviour falling out of the formulation and having to be bolted on. This is why Baraff's notes insist on it, and why Project 1 asks for the Dzhanibekov effect as a deliverable.

5 Rigid bodies in contact

- **An impulse applied at a point changes angular momentum:** $\Delta\mathbf{L} = \mathbf{r} \times \mathbf{j}$, where $\mathbf{r}$ is the vector from the centre of mass to the contact point. All of Module 06 carries over; this term is the only addition.
- **Effective mass is now direction- and position-dependent.** The denominator in the impulse formula gains terms involving $\mathbf{I}^{-1}$ and $\mathbf{r}$, so the same collision at the corner of a box and at the centre of its face produce different results — correctly.
- **Rolling without slipping is a constraint, not an emergent behaviour.** Friction alone produces rolling with some slip, which is often acceptable; true rolling contact requires an explicit constraint tying the contact point's velocity to zero.
- **Spinning friction** — resistance to rotation about the contact normal — is a separate effect and is usually omitted, which is why a spinning top or coin in most simulators never slows down on its own. Adding it requires a torsional friction constraint at each contact.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Baraff — Rigid Body Simulation I and II (free notes)	Sections D and G of the course pack. The state formulation of §1, the inertia tensor, and the argument for storing momentum. The primary reading.
Mirtich — Fast and Accurate Computation of Polyhedral Mass Properties (free)	The exact volume-integral method of §2 for computing inertia from a mesh. Complete code in the paper.
Dzhanibekov effect — NASA ISS footage	The physical phenomenon of §4, filmed in orbit. Watch it before implementing, then match it.
Erin Catto — Numerical Methods and rigid body GDC talks (free)	The practical side: quaternion integration, renormalisation, and what goes wrong in production.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Implement rigid body state and integration with quaternions. Verify that a body with no torque conserves angular momentum to floating-point tolerance over 100,000 steps.
- 2 Omit the quaternion renormalisation and plot the quaternion's magnitude over time. Report how long until the body visibly degenerates.
- 3 Compute the inertia tensor for a solid box and a solid sphere analytically and from your mesh code. They must agree.
- 4 Diagonalise the inertia tensor of an irregular mesh and visualise the principal axes. Confirm they are sensible.
- 5 **Reproduce the Dzhanibekov effect:** a body with three distinct principal moments, given a small perturbation from pure intermediate axis rotation. Plot angular momentum magnitude to confirm it is conserved through the flips.
- 6 Deliberately store angular velocity instead of angular momentum, without the gyroscopic term, and show that the flip does not occur.
- 7 Add the gyroscopic term to the angular velocity formulation and show the flip returns. Then compare its stability against the momentum formulation at large time steps.
- 8 Extend Module 06's collision response to rigid bodies with the $\mathbf{r} \times \mathbf{j}$ term. Verify that an off-centre impact imparts spin.

- 9 Build the Project 1 stack of ten boxes using full rigid body dynamics, and keep it standing for sixty seconds.
- 10 Implement a rolling constraint and compare against friction-only rolling. Measure the slip in each case.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 List the four components of rigid body state and their sizes.
- 2 Write the four equations of motion and say which quantity is derived rather than stored.
- 3 What is the inertia tensor, and why does it mean L and ω are not parallel?
- 4 How is the inertia tensor transformed to world space, and why store the inverse?
- 5 Give three rules for computing an inertia tensor correctly.
- 6 Compare three orientation representations and give the problem with each.
- 7 What happens if a quaternion is not renormalised each step?
- 8 State the intermediate axis theorem and say what is conserved during the flip.
- 9 Why does storing angular velocity instead of momentum break the flip, and what would have to be added?
- 10 How does collision response change for rigid bodies?